

-
- Nikhil Prakash, Natalie Shapira, Arnab Sen Sharma, Christoph Riedl, Yonatan Belinkov, Tamar Rott Shaham, David Bau, and Atticus Geiger. Language models use lookbacks to track beliefs. *arXiv:2505.14685*, 2025. URL <https://arxiv.org/abs/2505.14685>.
- Gautam Reddy. The mechanistic basis of data dependence and abrupt learning in an in-context classification task. In *The Twelfth International Conference on Learning Representations, ICLR 2024*, Vienna, Austria, 2024. OpenReview.net. URL <https://openreview.net/forum?id=aN4Jf6Cx69>.
- Noam Shazeer. Glu variants improve transformer. *arXiv:2002.05202*, 2020. URL <https://arxiv.org/abs/2002.05202>.
- Aaditya K. Singh, Ted Moskowitz, Felix Hill, Stephanie C. Y. Chan, and Andrew M. Saxe. What needs to go right for an induction head? A mechanistic study of in-context learning circuits and their formation. In *Forty-first International Conference on Machine Learning, ICML 2024*, Vienna, Austria, 2024. OpenReview.net. URL <https://openreview.net/forum?id=08rrXl71D5>.
- Curt Tigges, Michael Hanna, Qinan Yu, and Stella Biderman. LLM circuit analyses are consistent across training and scale. In Amir Globersons, Lester Mackey, Danielle Belgrave, Angela Fan, Ulrich Paquet, Jakub M. Tomczak, and Cheng Zhang (eds.), *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024*, Vancouver, BC, Canada, 2024. URL http://papers.nips.cc/paper_files/paper/2024/hash/47c7edadfee365b394b2a3bd416048da-Abstract-Conference.html.
- Asher Trockman, Hrayr Harutyunyan, J. Zico Kolter, Sanjiv Kumar, and Srinadh Bhojanapalli. Mimetic initialization helps state space models learn to recall. *arXiv:2410.11135*, 2024. URL <https://arxiv.org/abs/2410.11135>.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In Isabelle Guyon, Ulrike von Luxburg, Samy Bengio, Hanna M. Wallach, Rob Fergus, S. V. N. Vishwanathan, and Roman Garnett (eds.), *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017*, pp. 5998–6008, Long Beach, CA, USA, 2017. URL <https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html>.
- Roger Waleffe, Wonmin Byeon, Duncan Riach, Brandon Norick, Vijay Korthikanti, Tri Dao, Albert Gu, Ali Hatamizadeh, Sudhakar Singh, Deepak Narayanan, Garvit Kulshreshtha, Vartika Singh, Jared Casper, Jan Kautz, Mohammad Shoeybi, and Bryan Catanzaro. An empirical study of Mamba-based language models. *arXiv:2406.07887*, 2024. URL <https://arxiv.org/abs/2406.07887>.
- Boshi Wang, Xiang Yue, Yu Su, and Huan Sun. Grokking of implicit reasoning in transformers: A mechanistic journey to the edge of generalization. In Amir Globersons, Lester Mackey, Danielle Belgrave, Angela Fan, Ulrich Paquet, Jakub M. Tomczak, and Cheng Zhang (eds.), *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*, 2024. URL http://papers.nips.cc/paper_files/paper/2024/hash/ad217e0c7fecc71bdf48660ad6714b07-Abstract-Conference.html.
- Ke Alexander Wang, Jiaxin Shi, and Emily B. Fox. Test-time regression: a unifying framework for designing sequence models with associative memory. *arXiv:2501.12352*, 2025. URL <https://arxiv.org/abs/2501.12352>.
- Kaiyue Wen, Xingyu Dang, and Kaifeng Lyu. RNNs are not transformers (yet): The key bottleneck on in-context retrieval. *arXiv:2402.18510*, 2024. URL <https://arxiv.org/abs/2402.18510>.
- Sarah Wiegrefe, Oyvind Tafjord, Yonatan Belinkov, Hannaneh Hajishirzi, and Ashish Sabharwal. Answer, assemble, ace: Understanding how LMs answer multiple choice questions. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=6NNA0MxhCH>.

-
- Zhengxuan Wu, Atticus Geiger, Aryaman Arora, Jing Huang, Zheng Wang, Noah Goodman, Christopher Manning, and Christopher Potts. pyvene: A library for understanding and improving PyTorch models via interventions. In Kai-Wei Chang, Annie Lee, and Nazneen Rajani (eds.), *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 3: System Demonstrations)*, pp. 158–165, Mexico City, Mexico, June 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-demo.16. URL <https://aclanthology.org/2024.naacl-demo.16/>.
- Songlin Yang and Yu Zhang. Fla: A triton-based library for hardware-efficient implementations of linear attention mechanism, January 2024. URL <https://github.com/fla-org/flash-linear-attention>.
- Songlin Yang, Bailin Wang, Yu Zhang, Yikang Shen, and Yoon Kim. Parallelizing linear transformers with the delta rule over sequence length. In Amir Globersons, Lester Mackey, Danielle Belgrave, Angela Fan, Ulrich Paquet, Jakub M. Tomczak, and Cheng Zhang (eds.), *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024*, 2024. URL http://papers.nips.cc/paper_files/paper/2024/hash/d13a3eae72366e61dfdc7eea82eeb685-Abstract-Conference.html.
- Kayo Yin and Jacob Steinhardt. Which attention heads matter for in-context learning? *arXiv:2502.14010*, 2025. URL <https://arxiv.org/abs/2502.14010>.
- Wei Zhang and Bowen Zhou. Learning to update auto-associative memory in recurrent neural networks for improving sequence memorization. *arXiv:1709.06493*, 2017. URL <https://arxiv.org/abs/1709.06493>.

Appendix

Table of Contents

A	Model configurations	16
B	Formalisation of ATR	17
B.1	Additional details on ATR	17
C	Task hyperparameters	18
D	Additional experiments on AR and ATR	19
D.1	Attention needs position embeddings	19
D.2	Mamba’s solution to ATR does generalise	19
D.3	1-layer SSMs learn direct retrieval on AR and ATR	20
D.4	SSMs prefer layer 0 to perform AR	21
D.5	Rightmost sibling queries are trivial for all architectures	21